CIRP Journal of Manufacturing Science and Technology 35 (2021) 422–430

Contents lists available at ScienceDirect

CIRP Journal of Manufacturing Science and Technology

journal homepage: www.elsevier.com/locate/cirpj

Applied energy optimization of multi-robot systems through motion parameter tuning

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ARTICLE INFO

Article history:
Available online 10 August 2021

Keywords:
Energy optimization
Multi-robot coordination
Industrial robots
Robot stations
Motion parameters

ABSTRACT

In this paper, an optimization method for energy reduction of robot stations is presented, including an evaluation on an industrial robot station. The problem is formulated as a convex mixed integer nonlinear optimization problem, where the objective is to reduce the energy use by finding the optimal execution time and execution order of the robot motions. A simulation model of the station is used to find simplified energy models of the robot motions, that is used to solve the optimization problem. The optimal execution times of the robot motions are realized by tuning parameters in the robot control system. Different types of parameter settings are tested, such as reduced acceleration and velocity. The optimal parameter settings are then implemented in robot code in a real four robot welding station. The result shows a 12% reduction in energy use, without extending the cycle time of the station. A validation of the energy models used to solve the optimization problem is also made, by comparing them with real energy measurements.

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1 Introduction

Driven by an increased awareness of climate change as well as economic reasons, the effort on reducing the energy use of robot and automation systems in industry is rapidly increasing. Various approaches to the reduction of energy use of robot and automation systems exist, such as changing the hardware of robots and tools [1,2], optimizing the physical layout of robot stations [3], operation scheduling [4], turning off inactive machines [5] and trajectory optimization [6]. An overview of energy optimization methods for robots is published by Carabin et al. [7]. Other related literature are presented by Thiede et al. [8], that presents a concept for reducing the energy use of small and medium sized enterprises and Michalos et al. [9], that presents a practical case study for reducing the energy use (among other things) of an assembly line in an automotive factory.

Looking specifically at energy reduction of robot movements, one of the main challenges is to find the mechanical and electrical models of the robots, which is required in many energy optimization methods [10–12]. To find the parameters (masses, friction coefficients, capacitance, etc.), required to create such a model is however not always possible. As opposed to robots specifically designed for research purposes, this information is in general not available for industrial robots.

In theory, different types of system identification methods [13] could be used to acquire this information. Paes et al. [14] finds the energy optimal trajectory of an industrial robot by first performing system identification to find an energy model, and then perform the energy optimization. However, getting an accurate model can be challenging, especially if the robot has many degrees of freedom, and if both the mechanical and the electrical system are unknown. Furthermore, for an industrial robot used in production it might not be possible to execute the necessary excitation trajectories.

If it is not possible to create a complete energy model, another possibility is to use a simplified cost function that correlates to the energy use, e.g. mechanical power or acceleration [15–18]. Riazi et al. [15] proposes an optimization method that records an already existing robot motion and changes the velocity profile to create a more energy efficient one. Another simplified method is to use energy signatures to describe the energy use of the robot only as a function of the execution time [19,20]. Vergnano et al. [19] presents a method to optimize the energy use of a multi-robot system by using energy signatures. Bukata et al. [20] also use energy signatures and present an optimization method to find the energy optimal order and execution time of robot operations, as well as when to use power saving modes.

A second problem when working with industrial robots, is possible limitations in the control system. A common and effective

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https://doi.org/10.1016/j.cirpj.2021.07.012
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energy reduction approach is to directly change the trajectory of the robot [6,15,18,21], but in many robot control systems, that is not possible. In a robot control system, a movement is typically defined by a simple motion command that is used by the control system to generate the trajectory. It is often possible to set the velocity and accuracy of the movement, but not much insight is given into how the trajectory is created, and the options to affect it is limited. Attempts have been made to emulate the behaviour of a robot controller [22] and to use the built-in functionality of a robot controller in order to reduce the energy use of the robot [23–26]. Gleeson et al. [23] proposes a method to run trajectories on industrial robots by describing them by a set of robot instructions with different zone sizes. Garcia et al. [24] investigates which factors (speed, acceleration, payload etc.) that have the largest influence on the energy consumption of an industrial robot. Gadaleta et al. [25] presents a simulation tool for finding energy efficient robot movements by changing velocity and acceleration settings on the robot.

Another challenge when reducing the energy use of robots in a multi-robot station is that productivity normally is prioritized over energy reduction in industry. This fact, which is often overlooked in the literature, means that any optimization method used in practice needs to guarantee that the productivity of the station is not affected, neither by taking longer to finish than necessary, nor by risking a collision. In turn, this means that coordination between multiple robots needs to be considered and that changes to a real robot station preferably should be as small as possible.

The first contribution of this paper is to present a method for reducing the energy use of the robot movements in a robot station, that is possible to use in practice. It includes: using a simulation model to create simplified energy models of the robots; solving an energy optimization problem, which includes coordination of the robots and that result in optimal execution times of the robot movements; and implementing the results of the optimization by tuning of the motion parameters of the robots. The second contribution is to apply this method on a real industrial robot station, to show that the method is possible to use in practice and that it is able to reduce the real energy use of the station.

To the best of the authors knowledge, this combination of practical energy optimization through motion parameter tuning, multi-robot coordination and evaluation on a real industrial robot station, has not been presented before. Similar approaches have been presented, for example by Riazi et al. [27] and Glorieux et al. [18]. But their methods require direct modification of the robot trajectories, something that is not possible for the station considered in this paper. The work by Gadaleta [28] is also similar, but it does not include coordination of multiple robots, something that is needed for the station considered in this paper. The work by Vergnano et al. [19] does include coordination of multiple robots, but the results are not evaluated on a real robot station.

The paper is organized as follows: the energy optimization problem is presented in Section 2, and in Section 3 the real case study is described and optimized. The setup of the experiment on the real station is presented in Section 4, and the results of the experiment are shown in Section 5. Finally, some conclusions are given in Section 6.

2 Problem description

The problem of reducing the energy use of a robot station can be formulated as follows. In the station there is a set of robots $r \in R$. The station operates in cycles, and for every cycle, each robot needs to execute a set of operations before a deadline T. The operations are denoted by $i \in O$ and each operation has a starting time t_i and an execution time (duration) d_i. The execution times are constrained such that $d_i^{\min} \leq d_i \leq d_i^{\max}$, where $d_i^{\min}/d_i^{\max}$ is the minimum/maximum allowed execution time for operation i, and $d_i^{\min}$ is the default value of d_i.

Between the operations there are two different types of constraints. There are precedence constraints, which ensure that the execution of the operations follows a certain order. Let B be a set that contains every pair of operations (i,j) where i needs to be executed before j, such that

$$t_i + d_i \leq t_j \tag{1}$$

In the type of robot stations considered in this paper, it is known which robot that performs which operations, and the order in which the operations of one robot is executed is also fixed.

The other type of constraint ensures that two sequences of operations cannot be executed at the same time. The typical example for a robot station is when two robots need to use the same physical space, a so-called shared zone. In order to avoid collision, there is a constraint to ensure that one robot leaves before the other robot enters the shared zone. To clarify, two sequences of operations of two different robots that both require the same shared zone cannot be executed at the same time, and once a robot has started to execute its sequence of operations in the shared zone it cannot be interrupted by another robot. Let C be a set containing these constraints; every constraint $c \in C$ between two such sequences of operations can be expressed as

$$t_{c2} + d_{c2} \leq t_{c3} + Mz_c$$
$$t_{c4} + d_{c4} \leq t_{c1} + M(1 - z_c)$$

Here $c1$ denotes the first operation in the first sequence of operations, $c2$ denotes the last operation in the first sequence of operations, $c3$ denotes the first operation in the second sequence of operations and $c4$ denotes the last operation in the second sequence of operations. The variable z_c is a binary decision variable, determining which of the two sequences of operations that is executed first and M is a sufficiently large constant, e.g. T. When $z_c = 0$, the first sequence is executed before the second sequence and vice versa when $z_c = 1$. This technique is known as the Big M Method [29] and is used to model options in mixed integer problems. Finally, there is a constraint to ensure that every operation is finished before the deadline, $t_i + d_i \leq T$.

2.1 Optimization formulation

The goal is to reduce the total energy use E of the robots in the station. Specifically, the optimization formulation presented below is aimed at reducing the energy use of robot stations that are already used in production. That means that there are some limitations to what changes that are practically feasible to make to the existing robot programs, to not unnecessarily increase the risk

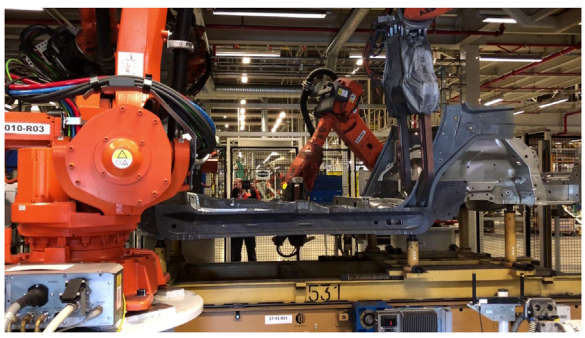


Fig. 1. A welding station from Volvo Cars.

of a collision, production delay, or other unforeseen consequences. For example, that is the reason why the operation order is partially fixed (through B) but also that the path of the robot trajectories are not allowed to be changed, only the timing.

As discussed in Section 1, to find a complete and accurate energy model of an industrial robot is challenging to do in practice. However, because of the constraint to not modify the path of the trajectories, the energy use of every operation i can be described by a function f_i^E, that only depends on the execution time d_i of the operation. Similar approaches have been used previously [19,30]. How to find these energy functions will be described in more detail in Section 3.2, but in short it is done as follows. Every operation is executed repeatedly with different motion parameter settings, e.g. velocity limits. Every such limit results in a specific energy use and execution time, and the energy function can then be created by fitting a function to this data. This allows the energy use of a robot to be described for every operation separately and without the need for a complete energy model of the robot. The complete optimization problem can be formulated as:

$$\min \quad \sum_{i \in O} (f_i^E(d_i) + \zeta t_i) \tag{2a}$$

$$\text{subject to :} \\ t_i + d_i \leq t_j, \quad (i,j) \in B \tag{2b}$$

$$\begin{cases} t_{c2} + d_{c2} \leq t_{c3} + M z_c \\ t_{c4} + d_{c4} \leq t_{c1} + M(1 - z_c) \end{cases} \quad c \in C \tag{2c}$$

$$t_i + d_i \leq T, \quad i \in O \tag{2d}$$

$$d_i^{\min} \leq d_i \leq d_i^{\max}, \quad i \in O \tag{2e}$$

$$0 \leq t_i, \quad i \in O \tag{2f}$$

$$z_c \in \{0,1\}, \quad c \in C \tag{2g}$$

The term ζt_i ensures that every operation is started as early as possible as long as it does not affect the energy use; ζ is chosen such that $\zeta t_i \ll f_i^E(d_i)$. In the optimization problem t_i, d_i and z_c are the decision variables.

2.2 Slack

Something that is not directly included in the optimization problem but is nevertheless important, is the amount of *slack* in the station. Slack can be used to quantify how much operations can be delayed without extending the total execution time of the station. In the literature several definitions exist [31,30], but in this paper the slack S of the station is defined as:

$$S := \sum_{i \in O} \begin{cases} T - (t_i + d_i), & \text{if } D_i = \emptyset \\ \min\limits_{j \in D_i} (t_j - (t_i + d_i)) & \text{otherwise} \end{cases} \tag{3}$$

where D_i is the set of operations that directly succeeds operation i. The slack is important for two reasons. Firstly, because there is, to a certain extent, a trade-off between slack and energy reduction. If the energy function $f_i^E(d_i)$ is decreasing while d_i is increasing (which is often the case), reducing the energy use also leads to reduced slack. In other words, the more slack there is a station, the more potential for energy reduction there is. For a station with zero slack it is not possible to reduce the energy use at all, using the optimization formulation presented above.

The slack is also important because it relates to how sensitive the station is to disturbances. If a station has a large amount of slack, the slack can "absorb" a disturbance occurring in one part of the station and prevent it from affecting the rest of the station. These two observations lead to the conclusion that there is a trade-off between robustness against disturbances and energy reduction. In [30] this trade-off is considered by using a multi objective optimization formulation. In this paper, disturbances are not included directly in the optimization problem, but is used to compare different solutions to it.

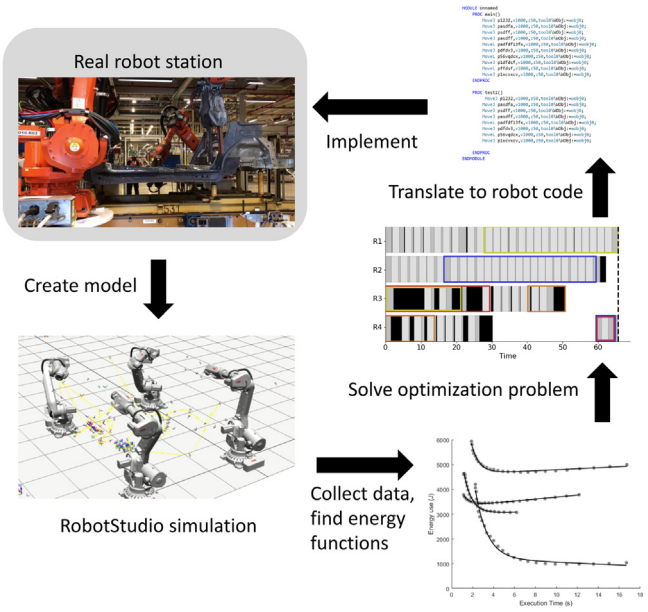


Fig. 2. An overview of the optimization workflow.

3 Case study

In this section the optimization formulation presented above is applied to a real industrial robot station. The station in question is a welding station from Volvo Cars in Gothenburg (Fig. 1). The station is part of a production line that produces chassis to cars and the cycle time is around 1 min. In the station there are four robots from ABB of the type IRB6640, each one equipped with a spot welding tool. In every cycle, each robot performs a series of spot welds on the car chassis. In the station there are four shared zones, preventing the robots from colliding.

3.1 Optimization workflow

An overview of the workflow for performing energy optimization of the station is presented in Fig. 2. A model of the station is created in the simulation software RobotStudio, which contains robot models, the robot code and zone constraints. From this model the data required to solve the optimization problem (2) are collected. Specifically, the energy functions f_i^E are created. The optimization problem (2) is then solved and the result is translated to robot code. Finally, the updated robot code is implemented in the real station.

For the purpose of energy optimization, the robot operations can be divided into two types. They are defined as: a *movement*

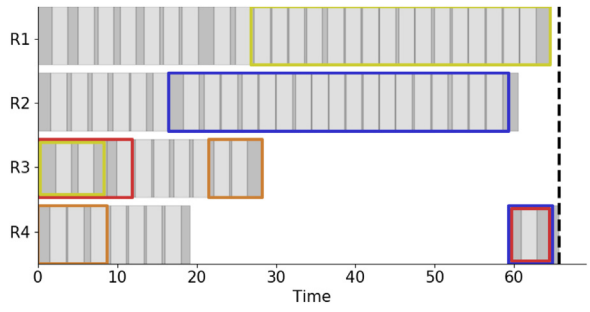


Fig. 3. A schedule of the unoptimized welding station, showing the four robots and the execution times of their respective operations. The welding operations are light gray, movement operations are grey, and the deadline is shown as a black dashed line. The colours indicate constraints between sequences of operations from different robots. Two sequences of operations with the same colour cannot be executed at the same time.

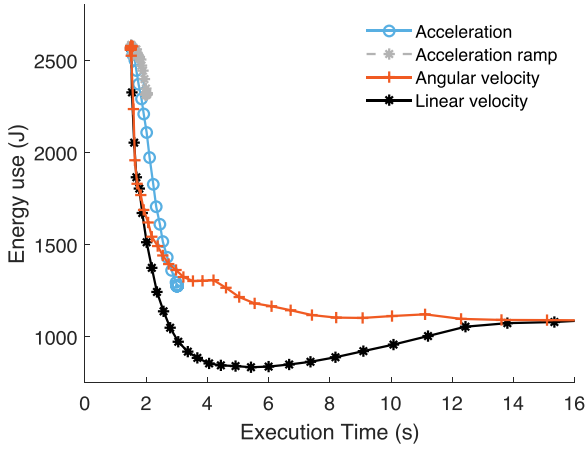


Fig. 4. The simulated energy use and execution times of an example operation, for the four different motion parameters.

operation is the set of movements required to go from one spot weld to the next, including closing of the welding tool, and a *welding operation* is when the robot is not moving and when the actual spot welding is performed. With this division the robot station contains roughly 70 operations of each type. A schedule of the simulation model of the unoptimized robot station is shown in Fig. 3.

Welding consumes a significant amount of energy [32,17], but reducing this energy is harder than for the movement operations. Modifying the power usage of a welding operation cannot be easily done without risking to affect the quality of the welds, which is not something that is worth the possible gain of reduced energy use. Therefore, reducing the energy use of the welding operations is not in the scope of this paper. Because of that, their energy functions can be excluded from the cost function (2a) and their execution times does not need to be optimization variables. Instead the execution time of the welding operations are approximated with a constant time of 2 s.

3.2 Energy functions

The energy functions of all of the movement operations are created by collecting energy data from the simulation model of the station. Data are collected by executing the robot code repeatedly with different motion parameter settings for each movement

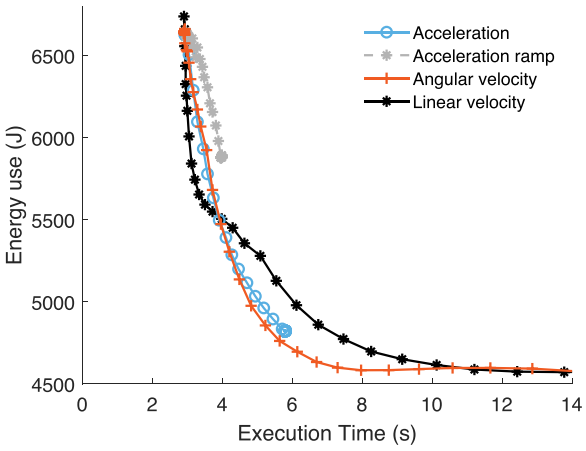


Fig. 5. The simulated energy use and execution times of another example operation, for the four different motion parameters.

operation, and the resulting execution time and energy use are recorded using a plug-in created with RobotStudio SDK (Software Development Kit). The motion parameters that are modified are (see [33] for reference):

- The maximum allowed linear velocity of the tool centre point (TCP) in mm/s that is set by adjusting the `v_tcp` component of the `speeddata` object for every move command. When collecting the energy data, the value in the first iteration was 2500 (the original setting), and for every new iteration the value was set to 90% of the value of the previous iteration. A total of 38 iterations were executed, which took around 45 min.
- The maximum allowed angular velocity of the TCP in degrees/s, that is set by adjusting the `v_ori` component of the `speeddata` object for every move command. When collecting the energy data, the value in the first iteration was 500 (the original setting), and for every new iteration the value was set to 90% of the value of the previous iteration. A total of 38 iterations were executed, which took around 35 min.
- The maximum acceleration (in percentage of the normal acceleration), that is set by changing `x` in the command `AccSet x,y`. When collecting the energy data, the value in the first iteration was 1 (the original setting), and for every new iteration the value was set to 90% of the value of the previous iteration. A total of 38 iteration were executed, which took around 30 min.
- The maximum acceleration ramp (in percentage of the normal acceleration ramp), that is set with the value `y` in the command above. When collecting the energy data, the value in the first iteration was 1 (the original setting), and for every new iteration the value was set to 90% of the value of the previous iteration. A total of 38 iterations were executed, which took around 30 min.

These four parameters are chosen because they are the ones that are easiest to implement in the robot code. Examples of the data collected for two example operations can be seen in Figs. 4 and 5 . It can be noted that the energy use more or less can be described by convex functions of the execution time. It can also be noted that there is a big difference between the two operations regarding which motion parameter that affects the energy use the most. It is not clear why, but it may depend on the configuration of the robot or the length of the motion.

With the collected data, $d_i^{\min}$ is determined for every operation and motion parameter as the minimum execution time, and $d_i^{\max}$ is determined as the execution time corresponding to the minimum energy use. For every operation and motion parameter the energy

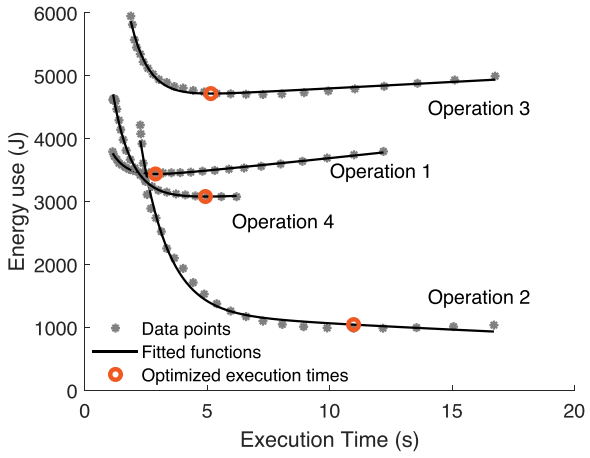


Fig. 6. The simulated energy use and execution time for some of the operations, together with the corresponding fitted functions, collected by limiting the linear velocity. The optimized execution time is also shown.

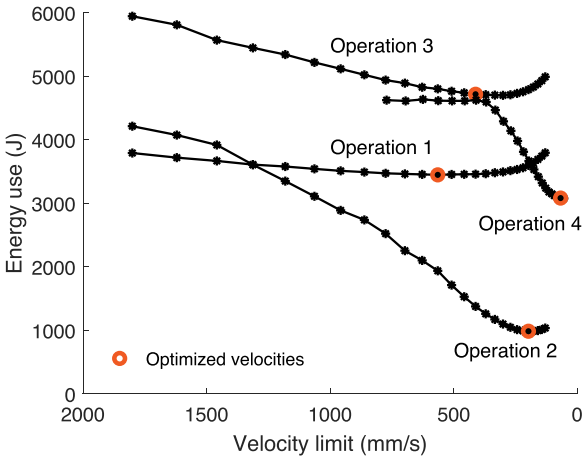


Fig. 8. The simulated energy use and velocity limits, collected by limiting the linear velocity. The optimized velocity limits are also shown.

Table 1
The optimized energy use and slack for every type of motion parameter.

	Lin. vel.	Ang. vel.	Acc.	Acc. ramp	Combination
Energy use (kJ)	83.3	83.8	85.7	90.5	82.4
Slack (s)	49.0	35.8	75.4	66.6	50.3

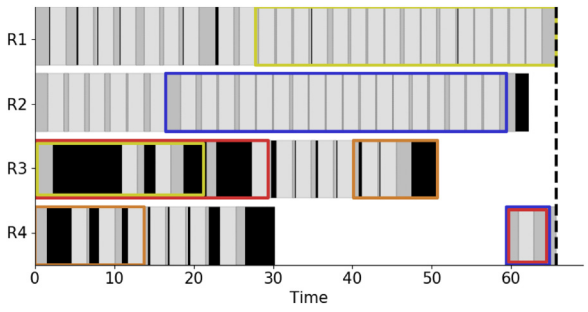


Fig. 7. A schedule of the optimized welding station. The black areas show the execution time extensions (compared to $d_i^{\min}$) after the optimization, otherwise the notations are the same as in Fig. 3.

function $f_i^E(d_i)$ is found by fitting a function to the recorded energy use and corresponding execution times. The function has the form $f_i^E(d_i) = \alpha_i \exp(\beta_i d_i) + \gamma_i \exp(\varepsilon_i d_i)$, where $\alpha_i, \beta_i, \gamma_i, \varepsilon_i$ are the fitting parameters and $\alpha_i, \gamma_i \geq 0$, so that the function is convex. Examples of the data collected by limiting the linear velocity together with the corresponding fitted functions for some of the operations can be seen in Fig. 6. As an example, the fitted parameters of operation 3 in Fig. 6 are $\alpha = 15,503, \beta = -1.356, \gamma = 4598, \varepsilon = 0.00421$.

This method of creating energy functions in not specific to RobotStudio or ABB robots. All that is needed is the ability to modify motion parameters and to record the energy use and execution times. It is therefore possible to do with other robot brands and simulation softwares, as long as the energy use is available. The advantage of using this type of energy functions, and the proposed simulation based method to find them, is that measurements from the real robots are not needed. Furthermore, the challenge of finding complete energy models of the robots are avoided by using the simplified energy functions, and these energy functions are reasonably accurate, because they are based on simulation models from the robot manufacturer.

3.3 Optimization result

With the energy functions, the optimization problem defined by (2) can be solved. It is solved once for every motion parameter and one time when a combination of energy functions of different motion parameters is used. This combination is chosen so that, for every operation, the energy function that has the smallest minimum value is used. The optimization problem is modelled using the optimization tool CasADi [34] and solved using IPOPT [35] and BONMIN [36]. Since the functions defining the constraints are linear (convex) and the cost function is convex, BONMIN solves the optimization problem exactly. The optimized energy use for the five instances are presented in Table 1, together with the slack. As a comparison, the original energy use if every operation is executed at $d_i^{\min}$ is 99.4 kJ and the slack is 83.9 s. The optimal energy use of all of the operations in the station if they are not constrained by T is 65.6 kJ; in that case the total execution time of the station becomes 114 s.

The result shows that the combination is best in terms of energy use, closely followed by linear velocity and angular velocity. Acceleration has the highest amount of slack but at the cost of a higher energy use. Linear velocity is the method that will be tested on the real station, it is simpler to implement than the combination and has more slack than the angular velocity. A schedule with the optimal zone allocations z_c^*, start times t_i^* and execution times d_i^* for the case with reduced linear velocity is shown in Fig. 7 (cf. Fig. 3).

The optimal solution is implemented in the robot code in the following way. In the original robot code, the order in which the zones are allocated is not fixed. It is decided on a first come first serve basis; the first operation that requests to enter the zone gets priority. The optimized zone allocations are the same as the original ones, so no modifications of the robot code are needed for the zone allocations. The same goes for the start time since every operation in the optimal solution should start as soon as the operations preceding it has finished executing, which already is the default behaviour in the robot control system. For the optimized execution times some modifications are required. Using the recorded data, the velocity limits that closest correspond to the optimized execution times are found, and these velocity limits are then implemented in the robot code. The optimized execution

times and the corresponding velocity limits can be seen for some of the operations in Figs. 6 and 8 .

4 Experimental setup

This section describes the setup of the experiment on the real robot station. The aim of the experiment is primarily to evaluate how much the energy use of the real station is reduced with the optimized settings compared to the original settings. A second purpose is to evaluate how accurate the energy models in the simulation software are.

In total the experiment consists of 44 cycles, alternating between original and optimized settings. Note that the velocity limits used in the experiment is not exactly the same as the optimized ones, e.g. the small adjustments for Robot 1 (see Fig. 7) are not included for practical reasons. Also note that during the experiment, the station is executed in "dry mode", which means that no actual welding is performed. Instead the robots are standing still for approximately the duration of a real welding operation.

A small overview of the electrical system of the robots is required in order to explain how the energy use is measured. The exact layout of the electrical system is hard to find, since it is not provided by the robot manufacturer, but a schematic view of the electrical system of a robot is presented in Fig. 9. In short, the robot is supplied with energy by a three-phase AC system. The AC is transformed to DC by a rectifier, which then supplies each of the six motors in the robot (one for each joint). Connected to the DC system are also a capacitor, that can store energy; a "bleeder", that can dissipate energy; and "auxiliary" devices, such as brakes of the robot. The brakes of the robot are normally closed, but during movement and for some time afterwards they are kept open by a constant supply of power. When a joint is decelerating, the electric motor is producing energy, and this energy is either used by any of the other motors or auxiliary devices, stored in the capacitor or dissipated by the bleeder.

Three different types of energy data are collected from the station during the experiment (see [37] for reference):

- *Simulated data:* The RobotStudio tool "signal analyzer" is used to record the power and energy use of the robots in the simulation model. There is no live connection to the robots; the power and energy use are based on a "nominal robot operating under typical conditions", and the energy use of the auxiliary devices is not included. This type of data is also what is used to find the energy functions.
- *Torque-based data:* By having a live connection to the robots it is possible to use the tool "signal analyzer online". With this tool, the power and energy use are based on torque data from the real robot. It is more accurate than the simulated data but also does not include the energy use of the auxiliary devices.
- *Measured data:* Current and voltage measurement equipment are connected to the three-phase AC system that supplies the robot cabinet. The total power and energy use are found by measuring both the current and voltage of every phase at a rate of 10,000 Hz. Note that the measured data is filtered with a Savitzky–Golay filter of order 1 and frame length 201 before it is used. Also note that the collected data includes the consumption of auxiliary devices.

The time to perform the experiment was limited and the measurement equipment could only be connected to one robot at a time. Therefore, the real measurement was only performed on Robot 4, and 3 cycles with original settings and 3 cycles with optimized settings were measured. Robot 4 was chosen, because it was one of the robots that had the most energy reduction. The linear velocity of the TCP is also recorded, using "signal analyzer online".

5 Result

The result is divided into two parts. The first part is a comparison between the three data sets and the second part is a comparison of the energy use between the original and optimized settings.

5.1 Comparison of energy data

In Fig. 10 a comparison is made between the simulated data, the torque based data and the measured data for Robot 4. Looking at the measured data, the power use is constantly higher compared to the other two data sets (around 600 W higher), and this is because of the consumption of the auxiliary devices. Looking specifically at the parts of the power use corresponding to the acceleration phase (e.g. around 18 s), the simulated data differs from the other two data sets. This is because simplified versions of the robot programs are used, which only includes the movements up to the welding locations; they do not include the exact behaviour of the welding procedure, such as closing of the welding tool for example. At the second acceleration phase (around 21 s), the simulated data follows the other two data sets more accurately, and this is because no welding occurs there. Looking specifically at the parts of the power use corresponding to the deceleration phase of the robot, the simulated and torque-base data show negative power use, and the measured data show a small decrease in power use. The behaviour of the measured data is because: during the deceleration phase, the negative power from the motors is used to power the auxiliary devices as well as to charge the capacitor, which means that less power is used from the AC supply. This is the case even a short period after the deceleration, when power instead is used from the capacitor.

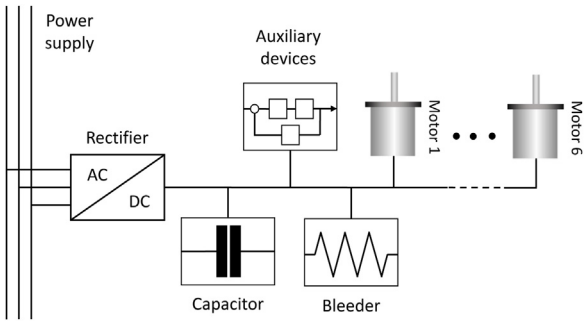


Fig. 9. A schematic view of the electrical system of the robots.

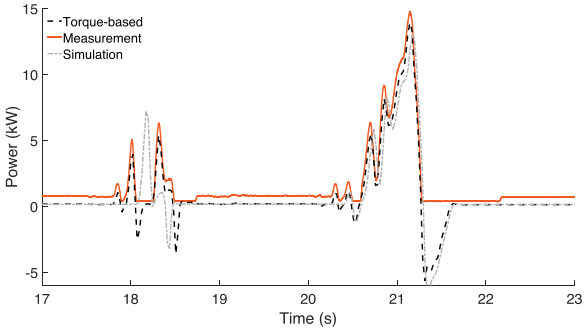


Fig. 10. A comparison between the simulated, torque-based and measured power use, for a part of the cycle of Robot 4.

Table 2
A comparison between the simulated, torque-based and measured power/energy use of Robot 4.

		Simulated	Torque-based	Measured (-auxiliary devices)	Measured
Peak power (kW)	Original	13.0	13.6	14.3	14.9
	Optimized	10.3	10.9	11.2	11.8
Energy use (kJ)	Original	29.4	30.2	30.5	70.4
	Optimized	24.6	23.8	25.0	64.8
Energy reduction (%)		16.3	21.0	18.0	8.0

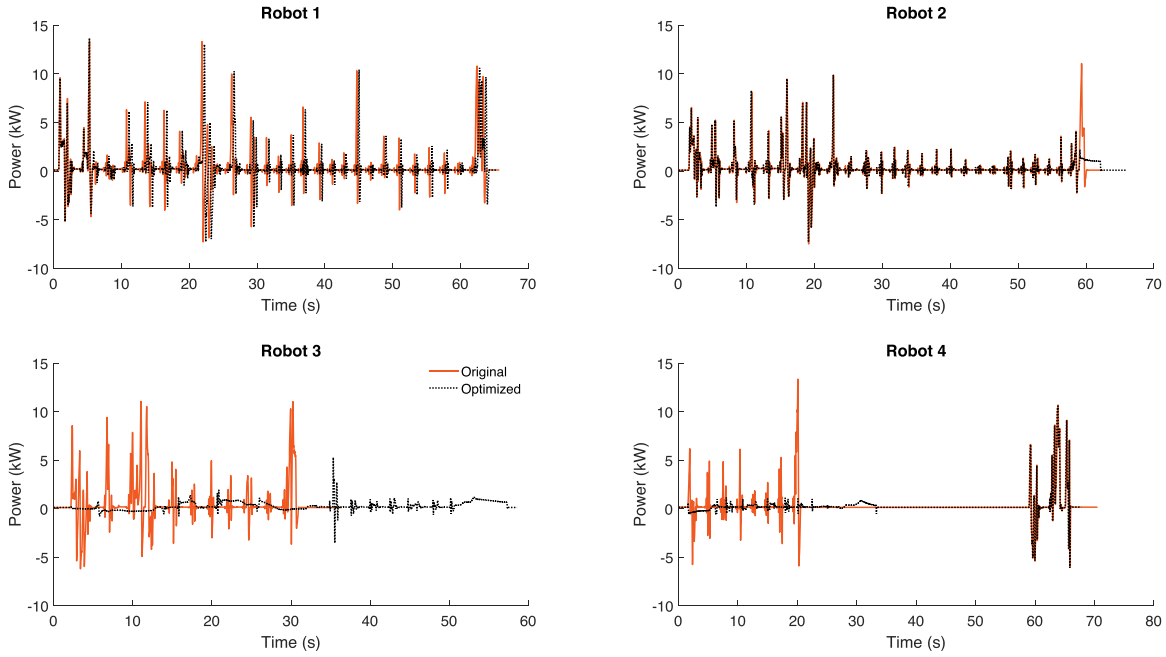


Fig. 11. The power use of the original and optimized settings for each one of the four robots.

In Table 2 the peak power, energy use and energy reduction are compared. The energy use for the simulated and torque-based data are calculated without including the negative parts of the power use. The peak power is calculated as the maximum power of one cycle. The measured data are shown both with and without the consumption of the auxiliary devices, assuming that the difference in idle consumption between the measured data and the torque-based data is due to the consumption of auxiliary devices. The result shows that, when disregarding the power use of the auxiliary device, the simulated and torque-based data slightly underestimates the measured data but is otherwise fairly accurate.

To conclude, although some differences can be seen between the simulated and torque-based data compared to the measured data in Fig. 10, the accuracy of the total energy use presented in Table 2 justifies the use of the simulated energy use during the optimization (which already has been presented), and the use of the torque-based energy use during calculations of energy savings (which will be presented in the next section).

5.2 Energy reduction

In this subsection the result of the whole station is presented. The values presented are the average of the torque-based data from

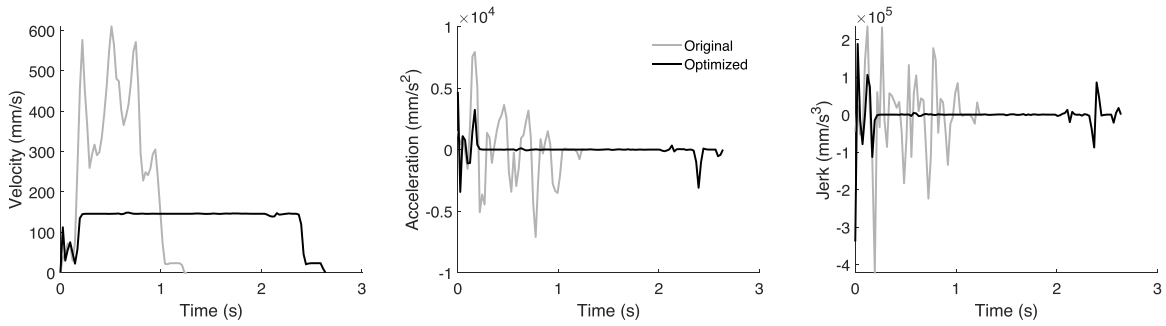


Fig. 12. The velocity, acceleration and jerk of one operation with the original and optimized settings.

Table 3
A comparison of the power and energy use of the whole station computed with the torque-based data.

Criteria	Robot	Original	Optimized	Reduction (%)
Energy use (kJ)	R1	40.4	40.7	−0.7
	R2	30.7	30.3	1.3
	R3	26.2	17.5	33.2
	R4	30.0	24.1	19.9
	Total	127.3	112.5	11.6
Peak power (kW)	R1	13.7	13.6	0.3
	R2	11.1	9.82	11.2
	R3	11.1	5.25	52.7
	R4	13.5	10.9	19.5
	Total	23.4	20.5	12.5
Average jerk (m/s^3)	R1	25.0	26.3	−5.2
	R2	28.6	27.3	4.3
	R3	25.2	6.31	75.0
	R4	11.7	8.20	29.9
	Total	90.5	68.1	24.7

the 44 cycles, which means that the auxiliary consumption is not included in the power and energy use. Fig. 11 shows the power use of the four robots, Fig. 12 shows a comparison of velocity, acceleration and jerk between the original and optimized settings for one operation, and Table 3 shows a comparison of the energy use, power use and jerk (mm/s^3) between the original and optimized settings.

Looking at Table 3, the biggest energy reduction are for Robot 3 and 4. This is as expected because they had more slack than the other robots before the optimization. During the optimization this slack has been used to lower the velocities of their operations and in turn reduce their energy use (see Fig. 7). Robot 2 only shows a slight reduction in energy use, and that is because it is only the velocity of its last operation that has been lowered. As already mentioned, no changes have been made to Robot 1 during the optimization. The slight increase in energy use that can be seen in Table 3 is within the margin of error for the measurements, and may for example depend on temperature variations. For the station as a whole, the result shows a reduction of 12% in energy use between the original and optimized settings. Note however that this value may be higher than what it would have been if real measurements had been used. Looking at Table 2, the torque-based data overestimates the energy reduction compared to the measured data for Robot 4. If this difference is compensated for when calculating the energy reduction of the whole station, the value would instead be 8%.

The peak power is of interest because it determines how powerful electrical components that must be used when setting up a robot station. Using less powerful components means that money can be saved. Looking at Fig. 11 the power peaks for Robot 3 and 4 are significantly lower after the optimization. For the station as a whole the peak power is reduced by 12%. Note that the peak power of the whole station is the maximum of the power use of the whole station and not the sum of the peak powers of the robots.

The average jerk is calculated as the mean of the absolute jerk of the TCP. Reducing the jerk increases the precision of the robot movements [38,39], but it is also believed that there is a correlation between jerk and wear and tear of the robot system. Reducing the wear and tear of the robot systems is something which is desired by industry, because it reduces the repair costs. Accurately determining the causes of wear and tear in a robot system is complex. One option is to look at the torque [40] but another option is to look at the jerk. By reducing the jerk, a robot will move smoother, and by that reducing the physical strain on the robot and the hoses and cables attached to it. For the optimized station the average jerk is reduced by 25%, compared to the original one. The reason why the jerk gets reduced, even though it is only the setting for linear velocity that is modified, can be seen in Fig. 12. Limiting the linear velocity leads to the velocity being constant for a large part of the execution time. In turn, this means that the acceleration and jerk are zero for that part. Hence, the average jerk is lower for the optimized settings than for the original ones.

6 Discussion and conclusion

In this paper a method for reducing the energy use of industrial robot stations by tuning of motion parameters has been presented. The aim has been to present a method that is simple enough to be usable for real industrial robot stations. Tests on a real robot station showed a reduction of the energy use, peak power and jerk. The result is promising, but it should be mentioned that the station used as a case study in this paper had an unusual large amount of slack. For a station with less amount of slack the improvements may be smaller. On the other hand, the optimization only used around half of the available slack in the station. Furthermore, looking at the shape of the energy functions, only small amounts of execution time extensions are required to achieve a significant reduction of energy use. So even for a station with small amounts of slack, it should be possible to reduce the energy use to some extent. The method has been tested on ABB robots using ABB's simulation software, but it is possible to use the method on other robot brands as well, if a simulation platform capable of estimating the power use of the robots is available.

It is also worth mentioning that the possibility for energy reduction was somewhat limited by the fact that the station was already used in production. Since any change to the real station would introduce a risk for collisions or other unforeseen consequences, the method used in this paper was aimed at being simple and unobtrusive enough to enable easy implementation in the real station. If the energy reduction would have been done at an earlier stage, during the initial design of the robot station, more freedom could have been given to the optimization problem. Things like how the welding operations are divided between the robots, in which order a robot performs its welding operations, the trajectories between the welding locations, and how to handle the shared zones could have been parts of the optimization problem, which likely would have resulted in more energy reduction.

Possible future work includes: automating the data collection and optimization process, to make the method easier to apply to a new station; including disturbances in the problem formulation; and trying the method on more (well-balanced) stations.

Conflict of interest

None declared.

Acknowledgements

The authors would like to thank Volvo Cars in Gothenburg for helping with the case study. This research was funded by Vinnova FFIthrough the project SmoothIT and the ITEA 3 project SPEAR. The support is gratefully acknowledged.

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